

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

**COURSE NAME: COMPUTER VISION with Open CV COURSE
CODE: DAS02**

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 40%

Instructions: Perform the experiment using the laboratory performance rubric Understanding of Concepts, Experimental Design, Use of Tools, Analysis of Results, Documentation.

1. Preprocessing Impact Analysis

Question: Design and perform an experiment to study the impact of preprocessing techniques such as noise removal and normalization on image quality and feature detection. Explain the theoretical concepts behind preprocessing methods and justify the selection of techniques used. Implement the preprocessing process using suitable software/tools, document the workflow and outputs systematically, and analyze how preprocessing influences feature detection accuracy and reliability.

Answer:

Preprocessing improves image quality before feature detection. Noise removal reduces unwanted pixels, while normalization adjusts pixel values to a standard range. Using Python and OpenCV, the image is filtered and normalized before feature detection. Preprocessing reduces false features and improves detection reliability.

Result: Preprocessing improves image quality and feature detection.

2. Image Representation Study

Question: Conduct an experiment to convert a given image into grayscale and binary representations. Explain the significance of different image representations in computer vision applications. Design the conversion procedure using appropriate tools, document the generated outputs clearly, and analyze how image representation affects edge detection and feature extraction performance.

Answer:

An RGB image is converted into grayscale and then binary form. Grayscale contains intensity values from 0–255, while binary contains only black and white pixels. These representations simplify image processing and improve processing speed.

Result: Grayscale and binary images are useful for edge and feature detection.

3. Edge Detection Comparison

Question: Design an experiment to compare Sobel and Canny edge detection techniques on the same input image. Explain the underlying concepts and operational principles of both methods. Apply the techniques using suitable tools/software, document the detected edges systematically, and analyze the comparative effectiveness of each method for different image conditions and applications.

Answer:

Sobel detects edges using horizontal and vertical gradients. Canny uses multiple stages including smoothing, gradient detection, thresholding, and edge tracking. Canny usually produces cleaner and thinner edges than Sobel.

Result: Canny is more accurate, while Sobel is simpler and faster.

4. Harris Corner Detection Analysis

Question: Perform Harris corner detection on images containing varying textures and patterns. Explain the concept of corner detection and its importance in computer vision tasks. Use appropriate parameters and tools to execute the experiment, document the

detected corner points clearly, and analyze the distribution, accuracy, and significance of the detected corners.

Answer:

Harris Corner Detection identifies points where image intensity changes strongly in multiple directions. These corners are useful for image matching and object recognition. Textured images usually produce more corners.

Result: Harris effectively detects important corner points.

5. Effect of Noise on Corner Detection

Question: Design an experiment to study the effect of noise on Harris corner detection performance. Explain the purpose of introducing noise and its influence on image features. Add noise systematically, perform corner detection using suitable tools, document observations for original and noisy images, and analyze the impact of noise along with possible improvement techniques.

Answer:

Noise creates unwanted intensity changes that can produce false corners. Gaussian or salt-and-pepper noise can be added to the image and Harris detection can be performed. Smoothing filters can reduce the effect of noise.

Result: Noise reduces corner detection accuracy.

6. Hough Transform for Shape Detection

Question: Conduct an experiment using the Hough Transform to detect lines or circles in an image. Explain the principles and significance of the Hough Transform in shape detection. Execute the detection process using appropriate tools, record the detected shapes and their parameters, and analyze the detection accuracy under different image conditions.

Answer:

Hough Transform is used to detect geometric shapes such as lines and circles. It converts image points into parameter space and identifies shapes based on accumulated votes. OpenCV provides Hough Line and Hough Circle functions.

Result: Hough Transform effectively detects lines and circles.

7. Feature Matching Using SIFT

Question: Perform feature extraction and matching between two images using the SIFT algorithm. Explain the concept of feature matching and scale-invariant features. Use suitable software/tools to extract and match features, document the matching key points systematically, and analyze the robustness of SIFT under variations in scale, rotation, and illumination.

Answer:

SIFT detects unique keypoints and creates descriptors for them. These descriptors are compared between two images to find matching points. SIFT is resistant to changes in scale and rotation.

Result: SIFT provides reliable feature matching.

8. PCA for Feature Reduction

Question: Design and implement an experiment using Principal Component Analysis (PCA) for dimensionality reduction of extracted image features. Explain the importance of feature reduction in machine learning and computer vision applications. Apply PCA using appropriate tools, document the feature dimensions before and after reduction, and analyze its impact on computational efficiency and feature representation quality.

Answer:

PCA reduces the number of features while preserving important information. It transforms original features into principal components. This reduces memory usage and computation time.

Result: PCA provides efficient feature representation with fewer dimensions.

9. Preprocessing vs Feature Extraction

Question: Conduct a comparative study of feature extraction performed with and without preprocessing. Explain the relationship between preprocessing and feature extraction performance. Execute both approaches using suitable tools, document the observations systematically, and analyze the differences in feature quality, detection accuracy, and overall system performance.

Answer:

Feature extraction is performed on an original image and a preprocessed image. Preprocessing removes noise and improves image quality. The preprocessed image generally produces cleaner and more reliable features.

Result: Proper preprocessing improves feature extraction performance.

10. Integrated Feature Detection Pipeline

Question: Design and implement a complete image processing pipeline consisting of preprocessing, edge detection, and feature extraction stages. Explain the purpose and workflow of the integrated pipeline. Execute each stage systematically using appropriate tools, document intermediate and final outputs clearly, and analyze the contribution and importance of each stage in achieving accurate feature detection results.

Answer:

The pipeline consists of:

Image → Preprocessing → Edge Detection → Feature Extraction

Preprocessing removes noise, edge detection finds boundaries, and feature extraction identifies important points. Each stage improves the final result.

Result: An integrated pipeline provides accurate and reliable feature detection.

11. Effect of Smoothing on Edge Detection

Question: Design and perform an experiment to study the impact of smoothing techniques on edge detection performance. Explain the theoretical purpose of smoothing in reducing noise and improving edge continuity. Apply smoothing and edge detection methods using appropriate tools, document outputs before and after smoothing systematically, and analyze how smoothing affects edge clarity, continuity, and noise suppression.

Answer:

Smoothing reduces noise before edge detection. Gaussian and median filters are commonly used. It reduces false edges and improves edge continuity. Excessive smoothing may blur real edges.

Result: Proper smoothing improves edge detection quality.

12. Gradient-Based Edge Detection Study

Question: Conduct an experiment using gradient-based edge detection methods to identify image boundaries. Explain the principles of gradient operators and their role in edge detection. Implement the method using suitable tools/software, record gradient values and detected edges systematically, and analyze edge strength, accuracy, and sensitivity to image variations.

Answer:

Gradient-based edge detection identifies areas with large changes in pixel intensity. Sobel and Prewitt operators calculate horizontal and vertical gradients. Strong gradients indicate strong edges.

Result: Gradient operators effectively detect image boundaries.

13. Prewitt vs Sobel Comparison

Question: Design a comparative study of Prewitt and Sobel edge detection operators. Explain the conceptual differences and applications of both methods. Execute both operators accurately on the same image using appropriate tools, document the outputs clearly, and analyze their performance in detecting edges under varying image conditions.

Answer:

Prewitt and Sobel use gradient masks to detect edges. Prewitt uses simple uniform weights, while Sobel gives more weight to center pixels and provides better noise resistance.

Result: Sobel generally produces more stable edges than Prewitt.

14. Effect of Image Scaling and Rotation on Feature Detection

Question: Perform an experiment to analyze how image scaling affects feature detection accuracy. Explain the importance of scale in computer vision and feature extraction tasks. Resize images to different scales, apply feature detection techniques, document detected features systematically, and analyze the impact of scaling on feature consistency and detection performance. Also evaluate the effect of image rotation on feature matching accuracy by rotating images at different angles and performing feature matching using a suitable algorithm.

Answer:

Images are resized and rotated by different amounts, then feature detection and matching are performed. Scaling can cause loss of small features, while rotation can affect matching. SIFT is robust to both scale and rotation changes.

Result: SIFT maintains good feature matching under different scales and rotations.

15. Threshold Effect in Edge Detection

Question: Design an experiment to study the influence of threshold values on edge detection quality. Explain the importance of thresholding in distinguishing meaningful

edges from noise. Apply edge detection using different threshold values, record the outputs systematically, and analyze how threshold selection affects edge continuity, accuracy, and noise sensitivity.

Answer:

Thresholding separates strong edges from weak edges and noise. Different threshold values are applied using Canny edge detection. Low thresholds detect more edges but also more noise, while high thresholds produce fewer and clearer edges.

Result: Proper threshold selection improves edge quality and accuracy.

16. Noise Removal Comparison

Question: Perform a comparative study of different noise removal techniques applied to images. Explain the objectives and principles of commonly used filtering methods. Implement multiple noise reduction filters using suitable tools, document the outputs clearly, and analyze the effectiveness of each technique in preserving image details while removing noise.

Answer:

Different filters such as **Gaussian, Median, and Bilateral filters** are applied to a noisy image. Gaussian smoothing reduces noise, Median filtering is effective for salt-and-pepper noise, and Bilateral filtering preserves edges.

Result: The best filter depends on the type of noise and required image details.

17. Feature Detection in Noisy Images

Question: Conduct an experiment to analyze feature detection performance on noisy images. Explain how image noise affects feature extraction and detection accuracy. Apply feature detection techniques on noisy images using appropriate tools, record detected features systematically, and analyze the impact of noise on feature reliability and stability.

Answer:

Noise creates unwanted intensity changes and may produce false features. Feature detection is performed on original and noisy images for comparison. Noise generally decreases the accuracy and stability of detected features.

Result: Noise reduces feature detection reliability.

18. Edge Detection under Illumination Changes

Question: Design an experiment to study the effect of illumination variation on edge detection performance. Explain the relationship between lighting conditions and edge extraction accuracy. Apply edge detection methods on images with varying illumination levels, document the outputs clearly, and analyze the robustness of the techniques under different lighting conditions.

Answer:

Different lighting conditions change pixel intensity and can affect edge detection. Edge detection is performed on images with different illumination levels. Poor or uneven lighting may create false or weak edges.

Result: Proper normalization and suitable edge detection improve performance under illumination changes.

19. Corner Detection Parameter Study

Question: Perform a parameter analysis study for Harris corner detection. Explain the significance of parameters in influencing corner detection accuracy. Execute the corner detection algorithm using different parameter values, record the detected corners systematically, and analyze how parameter variations affect detection sensitivity and performance.

Answer:

Harris corner detection depends on parameters such as block size, aperture size, and threshold. Different values are tested to observe changes in detected corners. Low thresholds detect more corners, while high thresholds detect only stronger corners.

Result: Proper parameter selection improves corner detection accuracy.

20. Hough Transform Parameter Analysis

Question: Conduct an experiment to evaluate the impact of parameter variations in the Hough Transform. Explain the role of parameters in shape detection accuracy.

Apply the Hough Transform using different parameter settings, document the detected shapes and outputs clearly, and analyze the sensitivity and reliability of the detection process.

Answer:

Hough Transform parameters control the detection of lines and circles. Changing threshold, minimum distance, and radius parameters affects the number of detected shapes. Incorrect values may produce false or missed detections.

Result: Proper parameter selection improves Hough Transform reliability.

21. Feature Matching Accuracy Evaluation

Question: Design an experiment to evaluate the accuracy of feature matching between images. Explain the objectives and significance of feature matching evaluation. Perform feature matching using suitable algorithms and tools, record matching points systematically, and analyze matching accuracy, false matches, and overall reliability.

Answer:

Feature matching identifies corresponding points between two images. Algorithms such as SIFT and ORB can be used. Correct matches and false matches are counted to evaluate accuracy.

Result: A higher number of correct matches indicates better matching performance.

22. PCA Component Selection Study

Question: Perform a study on the effect of varying the number of principal components in PCA. Explain the importance of component selection in dimensionality reduction. Execute PCA with different numbers of components, document feature dimensions and outputs clearly, and analyze the trade-off between dimensionality reduction and information preservation.

Answer:

PCA is performed using different numbers of components. Fewer components reduce computation but may lose information. More components preserve information but require more computation.

Result: The optimal number of components provides a balance between efficiency and information preservation.

23. Edge Detection vs Corner Detection

Question: Conduct a comparative analysis of edge detection and corner detection techniques on the same image. Explain the objectives and differences between both approaches. Apply suitable methods using appropriate tools, document the detected features systematically, and analyze the types of features captured and their relevance in computer vision applications.

Answer:

Edge detection identifies object boundaries, while corner detection identifies distinctive points where intensity changes in multiple directions. Canny can be used for edges and Harris for corners.

Result: Edges represent boundaries, while corners provide useful points for matching and recognition.

24. Multi-Stage Pipeline Analysis

Question: Design and implement a multi-stage image processing pipeline consisting of preprocessing, edge detection, and feature extraction. Explain the objectives and workflow of the pipeline. Execute each stage systematically using suitable tools, document outputs at every stage clearly, and analyze the contribution and effectiveness of each stage in improving overall system performance.

Answer:

The pipeline is:

Input → Preprocessing → Edge Detection → Feature Extraction

Preprocessing removes noise, edge detection identifies boundaries, and feature extraction finds important image points. Each stage improves the next stage.

Result: The multi-stage pipeline provides better and more reliable image analysis.

25. Feature Detection Density Analysis

Question: Perform an experiment to analyze the density of detected image features and its impact on performance. Explain the importance of feature density in computer vision tasks. Execute feature detection using appropriate tools, record the number and distribution of detected features systematically, and analyze the relationship between feature density, computational complexity, and detection accuracy.

Answer:

Feature density represents the number of detected features in an image. Highly textured images usually have more features and require more computation. Too few features may reduce matching accuracy.

Result: An appropriate feature density provides a balance between accuracy and computational cost.

26. Effect of Blurring on Feature Detection

Question: Conduct an experiment to study the influence of image blurring on feature detection. Explain the purpose of blurring and its effect on image details. Apply blur filters followed by feature detection techniques, document the outputs clearly, and analyze how blurring affects feature visibility, accuracy, and robustness.

Answer:

Blurring reduces image details and noise. Different blur levels are applied before feature detection. Slight blurring may remove noise, while excessive blurring can remove important features.

Result: Excessive blurring decreases feature detection accuracy.

27. PCA Visualization Study

Question: Perform a visualization study of features before and after applying PCA. Explain the significance of dimensionality reduction in feature analysis. Execute PCA using suitable tools, document visual outputs systematically, and analyze how PCA transforms and compresses feature representations while preserving important information.

Answer:

PCA transforms high-dimensional features into fewer principal components. Features can be visualized before and after PCA using 2D or 3D plots. PCA simplifies the feature representation while preserving major patterns.

Result: PCA makes high-dimensional features easier to visualize and analyze.

28. Feature Matching under Noise and Rotation

Question: Design an experiment to evaluate feature matching performance under combined noise and rotation conditions. Explain the challenges introduced by image transformations and noise. Apply noise and rotational transformations, perform feature matching using suitable tools, document the matching results clearly, and analyze the robustness and reliability of the matching technique.

Answer:

Noise can create false features, while rotation changes feature orientation. Noise and rotation are applied to an image and matching is performed using SIFT or ORB. The number of correct matches is compared with the original image.

Result: Robust algorithms such as SIFT can maintain good matching performance under noise and rotation.

29. End-to-End Vision System Experiment

Question: Design and implement a complete computer vision pipeline involving preprocessing, feature detection, feature matching, and PCA-based dimensionality reduction. Explain the objectives and workflow of the integrated system. Execute all stages systematically using appropriate tools, document outputs at each stage clearly, and analyze the overall system performance, limitations, and bottlenecks.

Answer:

The complete pipeline is:

Input → Preprocessing → Feature Detection → Feature Matching → PCA → Output

Preprocessing improves image quality, feature detection finds keypoints, feature matching identifies corresponding points, and PCA reduces feature dimensions.

Result: The integrated system provides efficient feature analysis, but performance depends on image quality, parameter selection, and computational resources.

Code of Conduct:

I K.L.V.S KARTHIK(192424177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.L.V.S KARTHIK(19242417)

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates	Good understanding with proper integration	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts

		multiple concepts/technologies	of most concepts		
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation